Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs in the Age of Transparent AI

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Problem Statement

The Core Question: How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2, RMSE)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability, statistical inference)

Research Objectives:

1. Compare ML and Statistical approaches across multiple problem domains
2. Quantify trade-offs between performance, cost, and interpretability
3. Provide evidence-based recommendations for model selection
4. Connect findings to Interpretable ML research and Transparent AI

Scope of the Study

Category	Count	
Problem Domains	4	Regression,
Datasets	13	
ML Models	5	Linear Regression, Random F
Statistical Models	5	OLS Regression, Logist
Total Comparisons	13	

Problem Domains & Datasets:

- **Regression:** California Housing, Diabetes, Wine Quality, Energy Consumption
- **Classification:** Iris, Wine Classification, Breast Cancer, Credit Default
- **Time Series:** Energy Consumption, Temperature, Sales Forecasting
- **Factorial Analysis:** Treatment Effects, Educational Outcomes, Agricultural Experiment, Drug Efficacy Study

Methodology: Regression

Statistical:

- OLS Regression (statsmodels)
- Provides p-values, confidence intervals
- Statistical inference capabilities

ML:

- Linear Regression (scikit-learn)
- Optimized for prediction
- Fast training and inference

Results:

Dataset	OLS R^2	ML LR R^2
California Housing	0.575	0.575
Diabetes	0.453	0.453
Wine Quality	0.787	0.787
Energy Consumption	0.811	0.970

Methodology: Classification

Statistical:

- Logistic Regression (statsmodels)
- Maximum Likelihood Estimation
- Provides odds ratios, p-values

ML:

- Random Forest (scikit-learn)
- Ensemble of decision trees
- Feature importance scores

Results:

Dataset	Logistic Acc.	RF Acc.
Iris	1.000	1.000
Wine Classification	0.956	0.982
Breast Cancer	0.956	0.965
Credit Default	0.999	0.999

Methodology: Factorial Analysis

Statistical:

- ANOVA (Analysis of Variance)
- Tests null hypothesis
- Provides F-statistics and p-values

ML:

- Support Vector Regression (SVM)
- RBF kernel for non-linear relationships
- Hyperparameter tuning via grid search

Results:

Dataset	ANOVA R^2	SVM R^2
Treatment Effects	0.417	0.513
Educational Outcomes	-0.002	0.284
Agricultural Experiment	0.464	0.745
Drug Efficacy Study	0.728	0.880

Methodology: Time Series

Statistical:

- ARIMA (AutoRegressive Integrated Moving Average)
- Auto-selection of (p, d, q)
- Interpretable coefficients
- Statistical diagnostics

Deep Learning:

- RNN/LSTM (PyTorch)
- Captures long-term dependencies
- Handles complex temporal patterns
- Requires more data and computation

Results:

Dataset	ARIMA R^2	LSTM R^2
Energy Consumption	0.450	0.890
Temperature	0.972	0.965
Sales Forecasting	0.972	0.969

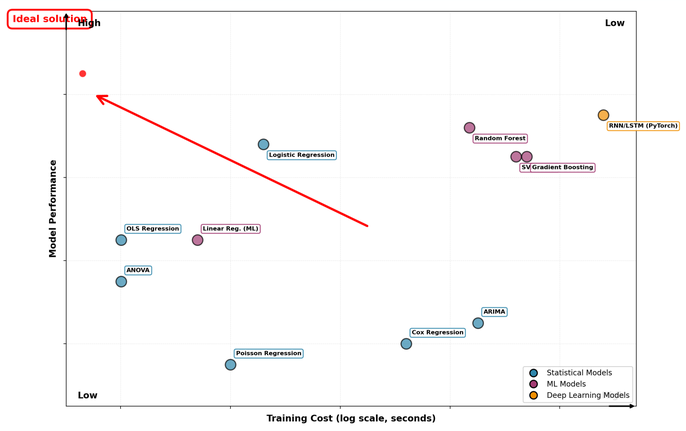
Overall Performance Summary

Metric	Statistical	ML/DL
Avg Performance (R^2/Acc)	0.547	0.818
Avg Training Time	0.037s	0.763s
Efficiency (Perf/Cost)	High	Medium
Interpretability	88.3/100	37.4/100

Key Statistics:

- Statistical models: 20-100x faster training
- ML/DL models: 1.5x higher average performance
- Interpretability gap: 2.4x difference

Performance vs Training Cost



Key Observations:

- Statistical models (blue): Low cost, moderate performance
- ML models (purple): Moderate cost, higher performance
- Deep Learning (orange): High cost, highest performance
- **Ideal solution:** Top-left corner (high performance, low cost)

Winners by Category

Category	Winner	Performance
Best Overall Accuracy	Random Forest (ML)	0.920
Fastest Training	OLS Regression (Stat)	0.0001s
Most Interpretable	OLS Regression (Stat)	100/100
Best Efficiency	OLS Regression (Stat)	6435.64
Best Time Series (Simple)	ARIMA (Stat)	$R^2 = 0.972$
Best Time Series (Complex)	LSTM (DL)	$R^2 = 0.890$

The Fundamental Trade-offs

	Cost	Interpreta
$P(\theta) = \mathbb{E}[\mathcal{M}(Y, \hat{Y})]$	$C(\theta) = T_{\text{train}}(\theta) + \lambda M(\theta)$	$I(\theta) =$ Expl
Metrics: R^2, Accuracy, RMSE	Training time + memory cost	Transparenc trust

The Optimization Problem:

$$\max_{\theta} \quad P(\theta) - \alpha C(\theta) + \beta I(\theta)$$

s.t. constraints

Efficiency Analysis

Ranking	Model	Efficiency
1	OLS Regression (Stat)	6435.64
2	ANOVA (Stat)	5445.54
3	Linear Regression (ML)	1297.41
4	Logistic Regression (Stat)	439.78
5	Poisson Regression (Stat)	349.65
...	...	...
Last	RNN/LSTM (DL)	0.36

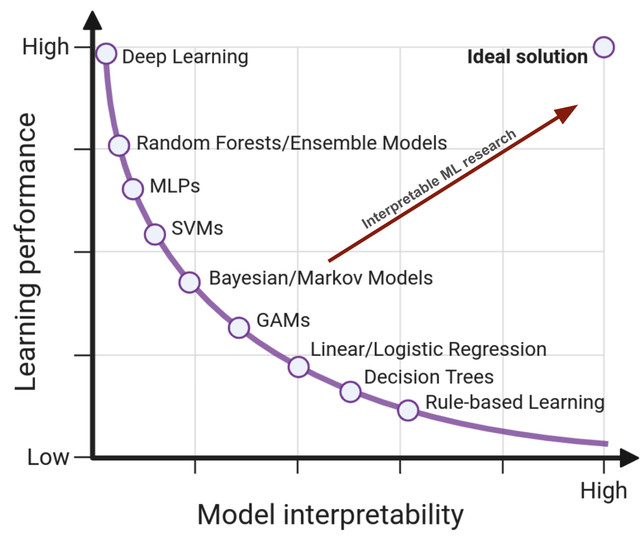
Statistical Models:

- 20-100x more efficient
- Best for real-time applications
- Low computational cost

ML/DL Models:

- Higher performance potential
- Better for complex patterns
- Suitable for batch processing

Performance vs Interpretability Trade-off



The Interpretability-Performance Curve:

- Traditional trade-off: Higher performance ⇔ Lower interpretability
- **Interpretable ML Research:** Moving toward the ideal solution
- Goal: Bridge the gap between statistical and ML approaches

When to Choose Statistical Models

Choose Statistical Models When:

- ✓ Interpretability is critical - Need p-values, confidence intervals
- ✓ Regulatory compliance - Transparent decision-making required
- ✓ Hypothesis testing - Understanding causal relationships
- ✓ Small to medium datasets - Optimal performance on limited data
- ✓ Real-time applications - Fast training and prediction needed
- ✓ Research and development - Need to understand underlying relationships

Best Models: OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox Regression

When to Choose ML/DL Models

Choose ML/DL Models When:

- ✓ Maximum accuracy is priority - Need best predictive performance
- ✓ Large, complex datasets - High-dimensional or non-linear patterns
- ✓ Production systems - Computational resources available
- ✓ Pattern discovery - Exploratory analysis